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45 Years with design methodology

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This is not an article! With this evident contradiction inspired by René Magritte's painting of a pipe, I will underline the special conditions I was given by the editor. The intention is that I shall review my own work and career, to articulate key ideas and to tell what I see as future challenges in my area. Therefore the use of 'I' is in a non-traditional form. The object of this article is the author's *Weltanschauung* concerning design and designing as it has developed over a 45 years period as teacher and researcher. Three dimensions are treated in an attempt to answer the following questions: how can we establish rigour and strong foundation for researching design? How to explain to industry what they are doing, and how to create industrial support? And what to tell the students about designing? I will focus upon the *dislocations* which have led to the development of the current state and what we see as a comprehensive school of designing. Details about established results can be found in the literature; I will focus upon the questions, thoughts, problems and beliefs behind the answers, and unsolved or non-clarified aspects. The article follows three lines of development, labelled Theory of Technical Systems, Engineering Design and Product Development, and our attempts to create a totality out of design philosophy, Domain Theory, Theory of Properties and our understanding of product development. Together they are the main part of 'our school', namely the foundation of the group 'Engineering Design and Product Development' at the Technical University of Denmark; the 'Copenhagen School' as our friends often refer to us. The conclusion attempts to balance in a joint model what I see as the role of design research in the worlds of teaching and practice, and where I see the challenges for the future.

Keywords: design methods; techniques and tools; design theory and research methodology; Theory of Technical Systems

1. Introduction

I became an MSc MechEng from the Technical University of Denmark (DTU) in 1965. I spent the last year of my education in the newly built campus in Lyngby, which was the result of an economical expansion and visionary plans. Staffing also needed expansion so to fill the need I was released from military service to join a project: the development of an artificial kidney system for Danish hospitals.

I belonged to a group surrounding Professor Vagn Aage Jeppesen, who established his chair at DTU in engineering design in 1952, founded upon a philosophy of design based upon creative

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thinking (Alger and Hayes 1964), product development (Asimow 1962), systematics (sparse signals picked up from Germany), and deep understanding of industrial practice. Jeppesen was very devoted to teaching systematic approaches and to bringing them to industry, as the first engineering design professor in Denmark. I joined the chair as assistant in 1966.

One of the unique ideas of Professor Jeppesen was the establishment of the Institute of Product Development in 1956, an independent foundation with the purpose to serve Danish industry and transfer knowledge and methods to practice. The institute was established with its own regular staff and we could alternate with this staff and bring them into our teaching, while we spent time in design and consulting activities. In the 1970–1980s I spent considerable time in industry, teaching design methodology and later as consultant on product development and organisational innovation.

The story told in this article follows me, the group of colleagues and the PhD students I have supervised, on a turbulent route of being an independent chair, being a laboratory, a department, a section belonging to Mechanical Engineering, and (currently) a section belonging to Management Engineering, with great fluctuations in staff volume. When I write *we* I refer to this group, sometimes very close to *pluralis majestatis* as a reality, but happily often with 5–10 colleagues as a team.

Our officially defined topic is ‘teaching mechanics’ with focus upon mechanical engineering education and courses on drawing, problem solving, Engineering Design and Product Development. In Sections 5.5–5.7, I will balance where our teaching activities today have brought us.

2. Theory of Technical Systems

The most influential knowledge-dislocation in my life was the meeting with Vladimir Hubka and his Theory of Technical Systems (TTS) (Hubka and Eder 1984). My education as a mechanical engineer had left me with an inhomogeneous world picture of ‘machines’ as related to machine elements, to elaborated ‘elementary cases’ of solid mechanics and to a broad spectrum of sciences. My design interests conflicted with these fractions: how to synthesise? Based upon what set of concepts and models of artefacts?

2.1. The beginning

In the establishing period in the 1960s our group was in a very free position, collecting material from various established groups in Europe and building up a rather comprehensive library. Our teaching duties were many, so not much written material was created.

Vladimir Hubka visited Denmark in the summer 1968 together with his family. He had a meeting with Vagn Aage Jeppesen, where Hubka told about his book manuscript on TTS and his ideas, and where he listened to Jeppesen’s endeavours, already crystallised into new courses and projects on engineering design. Few days later Hubka returned to DTU, now a fugitive because Soviet troops had invaded Prague and Hubka knew that he was seen as ‘politically unreliable’. Jeppesen hired him as design engineer in the Institute of Product Development (Figure 1).

Supported by study groups and practical product development projects (among these creation of an egg-sausage machine for industrial production of hard boiled eggs) we built up a joint TTS understanding, which did not disappear when Hubka in 1970 left Denmark for a position at ETH in Zurich, but grew in many directions and importance in the following years.

Hubka’s theory was an answer to my problematic world picture, stated above. In addition we felt that Hubka’s methods would support us in establishing a design methodology fit for our teaching at



Figure 1. Vagn Aage Jeppesen (1919–1975), Vladimir Hubka (1924–2005), and the author of this article Mogens Myrup Andreassen (1939–).

the university and satisfying industrial needs for design systematic. The theory defines the nature of technical artefacts as systems of organs (function carriers), systems of parts and articulation of a Theory of Properties, especially terminology concerning functions. As we shall see below this theory is also articulated or fitted to the synthesis of artefacts and leads to Hubka's model of designing. In Section 2.3 our terminology will be introduced systematically and throughout the article new terminology will be explained.

2.2. 'Design for X'-topics

In the above mentioned Institute of Product Development (IPU) a long series of industrial projects on development were performed in the 1970–1980s, building one-of-a-kind production and assembly machines. Hubka's book *Theorie der Maschinensysteme* from 1973 drew our attention to the so-called (Design for X) DFX-areas, especially Design for Assembly (DFA). Hubka had pointed to the nature and importance of how a technical system will be influenced in its design by the choice of manufacturing and assembling processes. Our projects gave us practical insight and we managed to find a logical structure of the topic, leading to two books: 'Design for Assembly' (Andreassen *et al.* 1988b) and 'Flexible Assembly Systems' (Andreassen and Ahm 1988c).

The core questions in design for assembly are:

How does the product design influence the operations in the assembly area?

What design principles should be followed?

Answering the questions call for a structured view upon both the design and the assembly activities. Three basic views from TTS were established:

- The *assembly process* is described as a system of activities, showing how the parts (operands) were brought into the assembly structure by assembly equipment or humans (operators).
- The *assembly equipment* is seen as a system, described by its functionalities and by the characteristics that are important for the assembly.
- The *product* is seen as a system of parts identified by the characteristics of these parts and their relations of importance for assembly.

This structure allowed us to crystallise and illustrate a long line of principles for DFA, linking statements on the product's characteristics to statements on the equipment's characteristics and pointing out what effects following the principle might have. Figure 2 shows the design degrees of freedom (A), which are applicable to both product and assembly system and pointing out the DFA-links (B). We see the articulation of a typical DFA principle. The illustration of four important relations (C) shows at the bottom level the fitting relations between parts and processes, which are important, but the alignment relations on higher levels (for instance creating familiarity relations

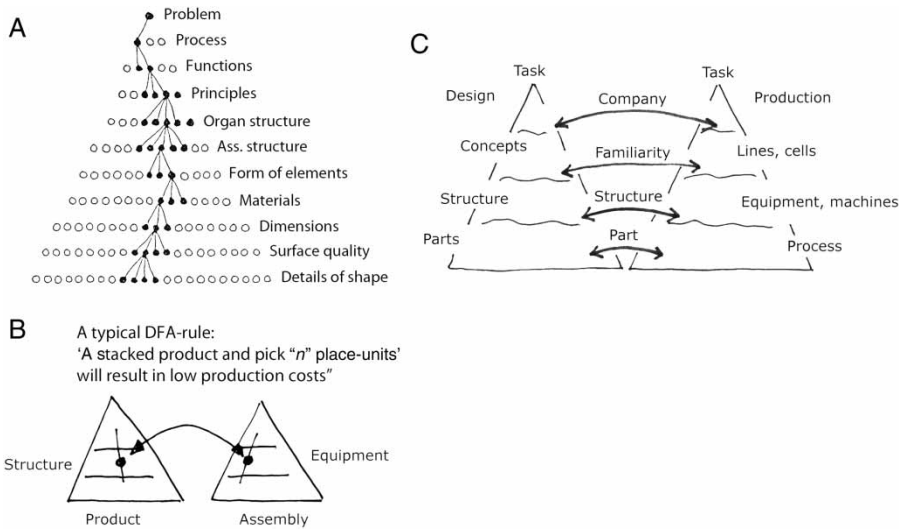


Figure 2. Design degrees of freedom model (A) utilised for articulation of a typical DFA principle (B) and general relations between a product and the production system (C) (Andreassen and Mortensen 1997a).

by modular thinking) are more powerful with respect to cost and time. We see the principles as 'conditionally valid', i.e. it is up to the designer to check if there are validity and effects to be reached in a certain situation.

The challenge of creating product variants, satisfying a spectrum of users' needs, but without raising the production complexity, and the challenge of creating assembly systems showing the necessary flexibility, were our focus for the second book. Here we laid the basis for modelling of products and machinery in such a way that we got insight into relations between the wanted, different functionalities and the product's way of being built up (German: Baustuktur), of importance to Design for Variety of the products and Design for Flexibility of the assembly system.

The internationally recognised results of our DFA activities were based upon a lucky combination of IPU staff designing and building assembly machinery and DTU researchers structuring and articulating the findings into textbooks. We have not since then found a similar research set-up of a 'build-experienced' research even if this type of research seems more powerful and convincing for product-related research than empirical approaches.

The work on DFA gave us a good foundation for later work on other DfX-areas and on creating formal models of artefacts and activity structures as we shall see in the following.

2.3. The dream of a 'Designer's Workbench'

Even if our group was not very active in CAD education and computer-related research, we took great interest in the question of creating a support system for designing; we called the dream a 'Designer's Workbench'. The dislocation were caused by our cooperation with Pedro Ferreirinha, a pupil of Hubka, co-operator in our informal society WDK (see Section 5.1), and the owner of an industrial consultancy in Switzerland. Ferreirinha's ideas were superior to ours; we had staff and finances.

The core question, we believed, was:

How to build a database for designing?

That is, how to make it possible to synthesise in a dialogue with a visual model on the computer screen. The idea here is that we shall go beyond the traditional CAD models and be able to reason

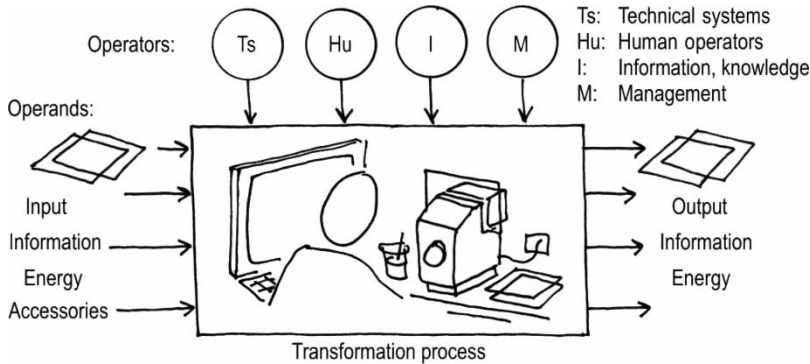


Figure 3. Hubka's general model of a transformation system (Hubka and Eder 1984), which delivers the basis for our workbench database (Tan 2010 after Hubka and Eder 1984).

about the products functional aspects. Hubka launched a very important illustration in his book on TTS, namely the general model of a transformation system (Figure 3). The basic idea is that any technical system or product is accompanied by a transformation, in which operands (material, energy, information and biological items) are transformed into a state, which satisfies the need. The effects necessary for the transformation will be supplied by the product, together with humans and the surroundings as operators.

Instead of transformation, I chose the word *technical activity*, defined as a single or sequence of transformations in which the product is utilised (as operator) or is transferred (as operand). The technical activity may be expanded to the total product life activity.

The transformation model relates important views upon a product: the technical activity (how to use the product), the effects or functions delivered by the product, and the product itself, which may both be seen as a system of organs and a system of parts, as mentioned before. I define organ as a functional element of a product, characterised by its actual function and composed of material areas (from the parts) and their interaction, which realise the function. In Section 2.7 is illustrated how we may look upon a product and choose to read its functional entities, organs, or its materialisation elements, parts, which are singular material entities.

Functions occupy the most important position in design methodology. By identification of the wanted function we, so to say, baptise the product, but as we shall see the reasoning about function is an intricate part of designing. I define function as an organ's, organism's or product's ability to create an active effect. The function depends upon stimuli, the organ's inherent physical, chemical or biological mechanisms and the state of the organ.

Ferreirinha *et al.* (1990) launched the word *chromosome* for a composed model consisting of four views or domains:

- An *activity view*, describing the technical activities and their relations, describing the use of the product, i.e. we simplified the model in Figure 2 by only looking upon the activities.
- A *function view*, describing the necessary functions or effects and their relations for establishing the activities, i.e. the product internal functions captured.
- An *organ view*, describing the organs and their interactions.
- A *parts view*, describing the parts and their assembly relations.

The Chromosome Model is shown in Figure 4 in its original form. Note the relations between the domains, shown in the figure. The model shall be seen as a metaphor of an ideal structure, where activities 'know' their organs and functions, and organs 'know' the parts, which realise the actual organ in a materialised form.

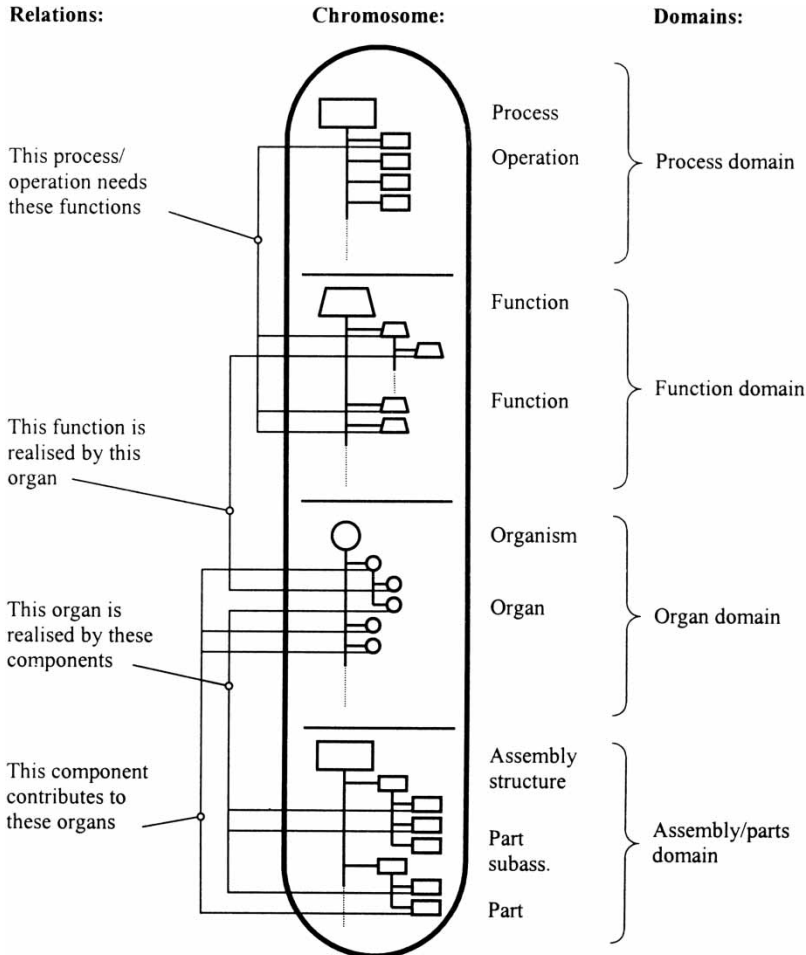


Figure 4. The original Chromosome Model with the later abandoned function domain (Jensen 1999 adopted from Andreassen 1980).

One of the great ideas of Hubka is the organ concept, namely to explain a product's functional behaviour by the abilities of organs (function carriers) to create effects, and their interactions. Pahl and Beitz (1996) propose a structure of functions as the entrance to finding solutions to each function. In German literature we see the efforts to understand the concept of functions, and we see very different considerations in a spectrum from function as behaviour and function as physical units in a product (German: Funktionseinheiten, i.e. similar to organs). In later applications we omitted the function domain, see Section 3.3.

We return to the important question of the nature of function and property throughout this article, because we have always seen it as a challenge to articulate the most productive concepts and find the answer to the question: what to tell the students?

The Chromosome Model may be used for organising data in a database, as shown in Figure 5, where each section defines the structure of activities, organs and parts. Any property related to these structures demand the establishment of a view model, i.e. a model where certain characteristics from the database are combined with contextual and situational characteristics forming a model, which can simulate the actual property. It might be a model of the manufacture, combining parts and production system characteristics for calculating costs, or it might be a model of

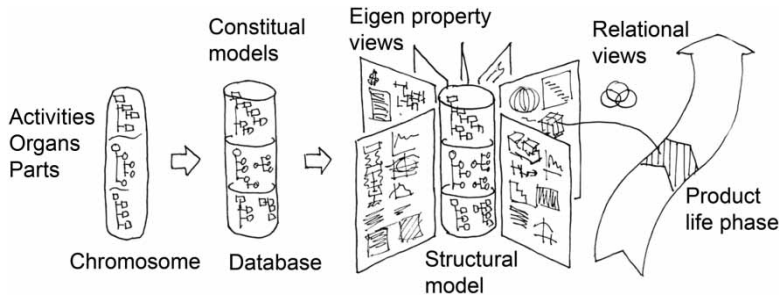


Figure 5. The ideal application of the Chromosome Model for creating a set of constitual models, from which so-called view models may be established (Jensen 1999, Andreasen 2007). Eigen properties and relational properties will be explained in Section 3.5.

product durability, combining the product's main body organ's characteristics with environmental influences.

2.4. Designing on a workbench

It was recognised (Andreasen 1990), that such a workbench should contain or be based upon:

- *Design language*, i.e. a vocabulary for thinking, reasoning, conceptualising and specifying solutions in all three domains, based upon semantics and syntax, and equally fitted for human reasoning and computer operations.
- *Design models*, i.e. models for structures of activities, organs and parts, carrying the specifications of these structures and allowing more or less formalised specification of relations inside and between the domains and of property statements of the entities.
- *Design operations*, i.e. methodologies for synthesising, composing, evaluating, modelling, simulating etc. for a gradual synthesis in all domains.

The core question of designing on such a workbench is:

How to support human design by computer operations?

In CAD systems one defines the artefact stepwise, from elementary, geometrical entities, or one imports chunks of structurally defined solutions. Our early imaginations were to use so-called *masters* for certain classes of design, which would consist of pre-defined models of frequent solutions but later we were able to propose new models as we shall see.

2.5. A pragmatic experiment: ALULIB

We were asked to join a Norwegian project ALULIB (Mortensen 1992) on developing a knowledge-based support tool for inspiring designers to use aluminium in their products. The basic idea was to establish a Chromosome Model for good product examples showing related organs and parts for each product and pragmatically adding information on product application and links to a database on aluminium technology.

In this way the user might search for products with similarities concerning application, context, functions and organs (hinges, covers, beams, bearings connections etc.), parts based upon manual search on formgiving (geometry), alloy choices, and finally details on aluminium technology related to the found products.

The database was established in an SQL relational database and data contributions were made by Scandinavian aluminium suppliers, to contain 120 products. 400 licenses were sold and very

positive reactions were obtained. We saw this as the first indicator that our workbench ideas were feasible. From a research point of view it was interesting to see how far one could get with a pragmatic, non-semantic database. Two factors added to the system's success: the use of visualisation and the use of domain relations: each product could be stripped down, showing its organs and parts, and one could find data on details.

2.6. *Realising the complexity*

Our results from ALULIB, our experiments with a computer-based system for design of bearing systems, CADOBs (Andreassen *et al.* 1988a), and the use of the design language TEKLA conceptualised by Ferreirinha *et al.* (1990), were the background for the formulation of a general specification and structure for a workbench, SMED (Andreassen *et al.* 1991), based on several publications on structuring of product data, product modelling, product development's functions in a workbench, and elaborations on a design language.

One of the central research questions to be answered for realising the dream about proper workbench operation was:

How to formalise the composition/decomposition of a design on a workbench?

It was the imagination of Karl-Henrik Svendsen that in a running design activity the task should be decomposed in accordance with the found concept, and goal formulations broken down accordingly, so that the next step, finding sub-solutions and composing them into a totality, could be supported by appropriate evaluations (Svendsen 1994). We simplified the situation by investigating pre-defined elements, i.e. organs, components and certain masters were pre-formulated.

The theory behind composing is what I called Hubka's First Law in my doctoral thesis (Andreassen 1980), namely the functions/means-pattern found in all products: 'In the hierarchy of functions, which contribute to the product's overall intentioned function there are causal relations, determined by the organs, which we chose to realise these functions'. When an organ is focused upon, and able to realise a given function, then this organ calls upon new functions to support or complement the organ's functions.

This causal pattern was first published by Hubka (1967) but did not get an explicit treatment before I wrote my thesis in 1980. The word *means* is used because actually transformations and user interactions can also be seen as creators of functions.

The research was confronted by complexity problems, forcing us to use simplified design experiments. The research showed us fundamental limitations in our research and in design theories (for instance missing goal decomposition and balancing theory) concerning the possibility to reach computer support in handling organ relations and specification breakdown. But we found ways of pragmatically modelling structures and utilising the function/means-pattern as design support (Svendsen and Hansen 1993).

2.7. *Clarification of organ structures*

The daring postulate in our workbench research is the idea that designing shall be performed as reasoning about organs and building organ structures, before or parallel to the definitions of the parts and their assembly relations. A challenge in designing is to capture the reasoning and intention behind the choice of organs:

How can design intent be linked to a structural product model?

In designing it is a very delicate problem, known from research, learning and practice, to come from an organ principle to the materialisation in the parts domain. Therefore it might be attractive to be able to alternate between organ reasoning and part reasoning in a computer support system.

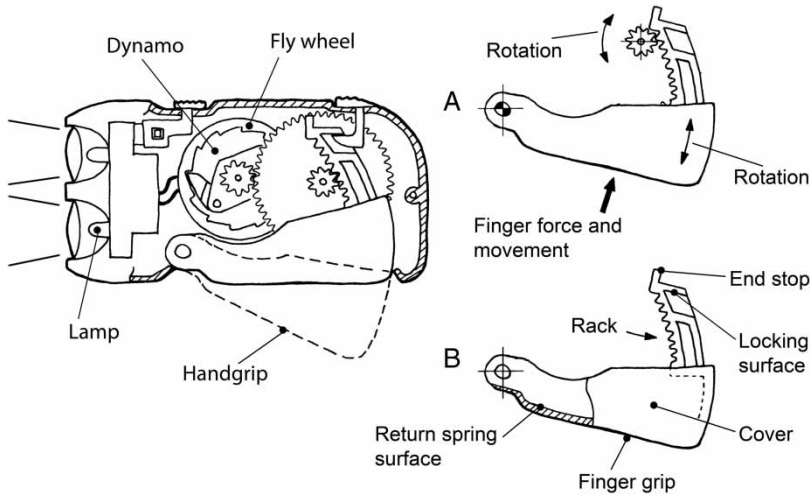


Figure 6. A human powered toy torch used to illustrate a system from an organ and parts viewpoint. The force/movement organ is shown (A) and the hand grip part's relation to this organ and other organs explain this part's tasks (B).

Research was performed by Jensen (1999) investigating the nature of organs, their interactions, mechanisms in creating functions or effects, and the design reasoning related to designing based upon organ thinking. Theoretical and experimental approach using paper-based prototyping and experiments with experienced designers showed the following:

- The organs are the carriers of behaviour (functions, properties) and therefore the core knowledge elements in designing.
- Organs are composed of what Jensen call *wirk* -elements, i.e. the active or activated form-elements.
- In the parts structure we find these form-elements as the important features of parts, to be respected in the part structure.

The prefix *wirk*- is taken from German, meaning action- or effect-. Organs deliver *Wirkung* or effect (*Auswirkung*), which act on other organs (*Einwirkung*). This pattern of relations is our concern when composing a solution. In the example in Figure 6 the handgrip takes up effect (force and movement) from the fingers and transfers it to the rack and pinion to create rotation.

Understanding an organ structure means to understand the state transitions and *wirk*-reactions propagating through the structure. A condition for applying workbench design seems to be that we operate with organ units, i.e. knowledge units clustering function and organ insight. The research of Jensen showed the principal possibility of formulating generic information structure for organ units, but it becomes surprisingly complex and semantically, properties seem difficult to obtain.

Even if our attempt to create a design language and develop support facilities in the transition between organ and part domains was hitting a barrier of complexity, – and we had to give up our workbench dream, we were quite content with our results. Many contribution to TTS in the form of models and methods were created, so that we were ready to struggle with a wide span of research topics from detailed design to product family configuration, modularisation and platform thinking.

2.8. A new direction: modularisation

Our research efforts related to the workbench concept had specific characteristics: it was derived from our own ideas about designing, the students were allowed to theorise, and dialogue with

industrialists was, so to say, the validation tool. Theorising without a foundation of empirical research is today seen as a questionable approach. We felt that we worked with innovative ideas and that satisfied us.

Radical dislocations occurred by the end of the millennium in changing industrial conditions due to what with a buzzword is called 'globalisation', enforced competitions and environmental concerns. Balanced by McAloone *et al.* (2007) we see the industrial challenges translated into the following list of necessary new engineering responsibilities:

- Enhanced quality efforts.
- Customer-oriented quality, values and perceptions.
- Environmental concerns and demand for sustainable solutions.
- Exploding design complexity due to technologies, multi-product development, customisation, legislation and product life concerns.
- Mass-customisation, platform thinking.
- Multi-disciplinary product conceptualisation.
- Globalisation of markets, supply, technology and customers.
- Necessary dynamic innovation of products and organisations.
- Handling of knowledge and competences.

This development caused fundamental changes in our design philosophy, as we shall see, concerning design for environment, conceptualisation, product life thinking and multi-product development. Here we will follow up on the effects on our TTS research: the modularisation research.

Mass production confronted by product individualisation and more precise need satisfaction leads to wishes for configurable products, for instance by combinatory module systems. A module is a product entity, which from a function or organ point of view has distinct function and requested properties, but at the same time such interfaces and interactions with other entities that you can see it as a building block in the parts structure. It is evident that research on modularisation is strongly supported by TTS, and as we shall see in the following, the Domain Theory.

Mortensen (2000) created in his thesis a Generic Design Model System, covering part structure language, modelling of the product's-related activities over the life cycle, and the establishment of view models (Figure 5). So the system is able to cope with characteristics and properties in all three domains.

Modularisation is aiming at creating variety seen from the customer's viewpoint, whilst at the same time showing kinship or commonality between module variants, and such structural properties, that it reduces the complexity in the company's operations.

Figure 7 shows the important aspects of modularisation and platform thinking: (A) the product's modular architecture created in the design phase of the product life respects the desired variety, creates kinship or commonality for efficient use of resources, and reduces effects of complexity in all actual operations. (B) A platform may be seen as alignment of architectures, normally product- and production architectures, but generally architecture related to any relevant life cycle activity and related to knowledge. (C) The reason for creating modular design and platforms is the alignment of the company's activities to harvest benefits in certain life phase activities, here design, production, distribution and disposal.

So we have here a situation parallel to the DFX-question:

How can we create a modular structure with customer benefits, which at the same time is fitted to the DFX-areas?

Modularisation is a very composed area:

- (1) *Designing modules* as a complex function and part design operation is a detailed design operation.

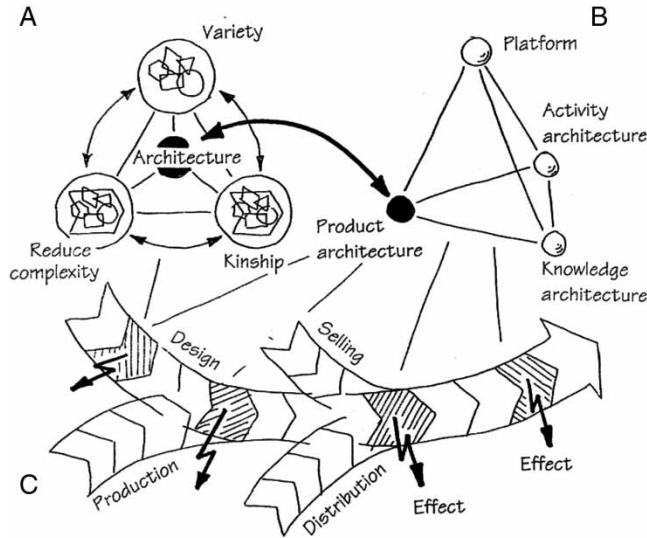


Figure 7. Product architecture's relation to variety, kinship and complexity plus rationalisation effects in different DFX-areas by a platform controlled alignment (Andreasen *et al.* 2001).

- (2) *Creating modular architecture* determining modules, interactions and interfaces allowing configuration of modular products includes two distinct different tasks:
 - to interpret and decide about the product family content and range from a market point of view
 - to create an architecture, containing a minimum of modules for creating total variety
- (3) *Harvesting benefits* of modularisation in areas where the effects can be seen, for instance in design, production, supply, re-use etc., by creating optimal conditions for these areas. The fitting can be called *alignment*; the resulting aligned structures are *platforms*.

Figure 7 shows in abstract form the modularisation's relation to customers, company internal operations, product life thinking and creation of aligned platforms.

2.9. Applied modularisation

Modularisation is a rewarding area for abstract research considerations; the proof is in the application. It is very important to note that modularity is a *relational property*; it has no meaning to analyse and describe a product's seemingly modular structure unless its fit to a certain company area is known: how benefits of modularisation are created. Miller (2001) created configurative modularisation for developing complex medical power plants, and since Professor Niels Henrik Mortensen took over this research area, a long line of candidate projects and PhD-projects have focused upon creating product families, modular architecture, architecture management and platform thinking in Danish companies. In this way substantial research contribution to modularisation theory has been created. Our state-of-the-art is described in Harlou (2006), Kvist (2009), Nielsen (2009) and Pedersen (2010).

2.10. TTS retrospectively

In our group we see TTS as the most influential basic theory. The lesson learned here is, in my opinion, that PhD-research based upon picking concepts and ideas from different authors and

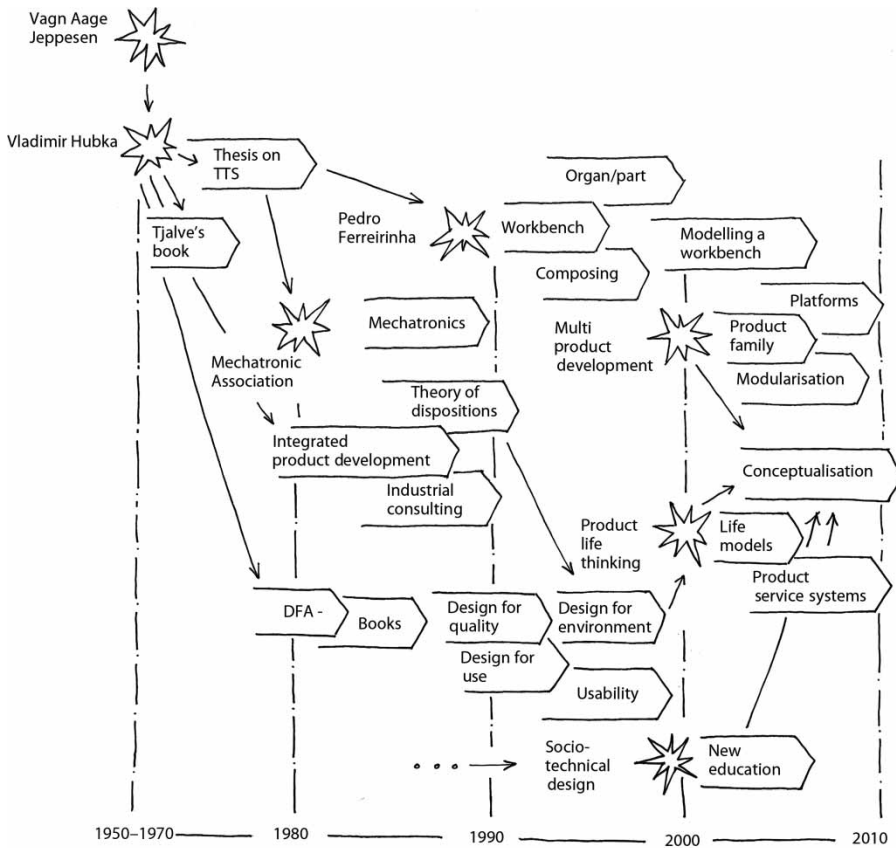


Figure 8. An attempt to illustrate TTS's influences, my dislocations and the results and their mutual influences, presented throughout this article.

trying to link them into a kind of foundation leads to fragmented and not easily understood and argued research. We have fortunately been able to work in many different directions based upon TTS as unifying theory.

Figure 8 summarises the influences of the TTS injection, pointing to engineering design, workbench design, DFX-developments and product life thinking. The results in these areas are presented in the following.

3. Engineering design

As mentioned, our group was established by Professor Jeppesen, with a substantial recruitment of senior research staff in the late sixties. He visited USA on a Marshall programme and returned inspired by American design education and industrial power. His teaching contained a design process model and methods inspired from Asimow (1962), Alger and Hayes (1964), Harrisberger (1966), and McKim (1972) with focus upon creative and systematic methods of synthesis. His pedagogical approach was furthermore based upon the philosophy, that the designer should be 'confronted with the need situation' and as result be able to crystallise ideas for need satisfaction.

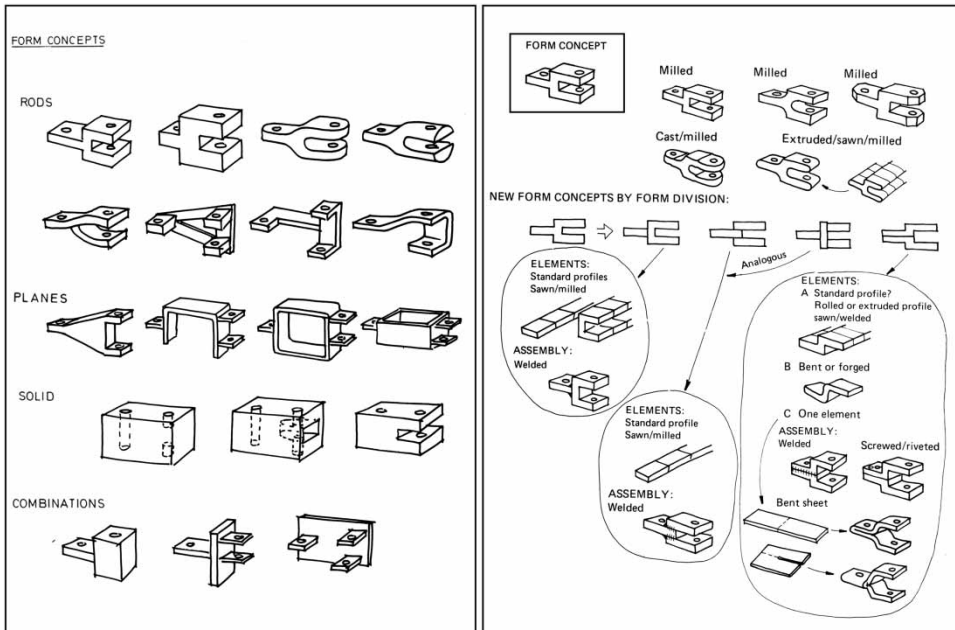


Figure 9. Often-quoted illustrations from Tjalve's (1979) book showing form concepts based upon form variation.

3.1. A unique textbook

Jeppesen's view on sketching ability as fundamental for designing, and Hubka's injection to our group on systematic and methodical design was integrated into Tjalve's (1979) textbook 'Systematic formgiving of industrial products'. Tjalve was distinctly graphically (Figure 9) and creatively gifted and based the book upon his experiences and examples from industrial projects at IPU and the recognised needs from our education.

The importance of graphical methods in engineering was not generally recognised. The English editor insisted on calling the book 'A Short Course in Industrial Design', and a German reviewer of Hubka's translation of the book into German only focused upon what he saw as the relation between engineering and industrial design.

Tjalve's book distinguishes itself as a practice-oriented and highly inspiring textbook, and appears as a pedagogical elaboration on Hubka's theories. Its simple, yet clearly focused design synthesis procedure, its balance of systematic and creative methods, the philosophy of product life concerns and product usability, and the power of his graphically supported methods, makes this book a classical work on designing.

3.2. Graphical modelling

One of our teaching topics is technical drawing and sketching. Based upon TTS, our design experiences and studies of books like McKim (1972) we established a new course 'Graphical modelling' about 1974. The basic ideas behind the course were (Tjalve *et al.* 1979):

- To relate all types of drawings and sketches to be learned to design situations, focusing upon the student's ability to apply graphical models for definition, analysis, communication and synthesis. Therefore all exercises were design task oriented.

- To skip topics traditionally combined with a drawing course: report writing, creative methods, artistic sketching, etc.
- To supply the students with a rich vocabulary of drawing types, to be fitted to situational characteristics like modelled properties and the receivers' background and application of the drawing.

It was a supportive situation for the teaching that the course could run over two weeks, 8 h per day, with no other obligations interfering. The students appreciated the learning and commented: 'You are fooling us to work hard', as a comment to the inspiring and varying tasks.

The course gave us a good starting point for later research on design language and modelling, especially the use of graphics in industry, and supplied staff with a graphical ability, which was soon recognisable in our publications and presentations at conferences.

3.3. A thesis on methods

Defending a thesis was a necessary condition for my further career as researcher, so we arranged a period of part-time work to create space for my research. I defended my thesis at the University of Lund, Sweden (Andreassen 1980). The research question in my thesis may be articulated as:

What concepts shall be used for reasoning about the synthesis and structuring of mechanical products?

You may say that the answers were already given by Hubka, but I had joined many discussions with Hubka and knew where the concepts may not be the best and where deeper clarification was necessary. I also found that some reformulations were necessary for being able to give the area a more comprehensive and above all a pedagogic formulation.

The thesis contributed to systematic and methodical understanding of machine design and showed how system theory, models and methods could be 'used four times' by viewing the product and its use activities as systems: an activity system, function system, organ system and part system. These four views you find in the many years later created Chromosome Model (see Section 2.3). I called these structural views *domains*. Each domain shall be seen as a system and in each domain the system's attributes should be distinguished as structural *characteristics*, which define or specify the system, and behavioural *properties*. Because the set of words are seen as synonyms in daily talk, the choice is arbitrary, but the distinction very important. My terminology was taken up by Weber (2005) in his CPM/PDD-theory and is gaining popularity in design science. It reduces Hubka's external, internal and design properties to two classes: characteristics and properties. Figure 10 shows our choice of attribute terminology. Note that we see functions as a class of behavioural properties, distinguished by being active as defined in Section 2.3.

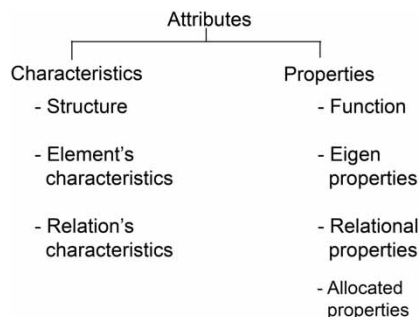


Figure 10. Structural characteristics and behavioural properties, broken down into classes. Definitions will follow throughout the article.

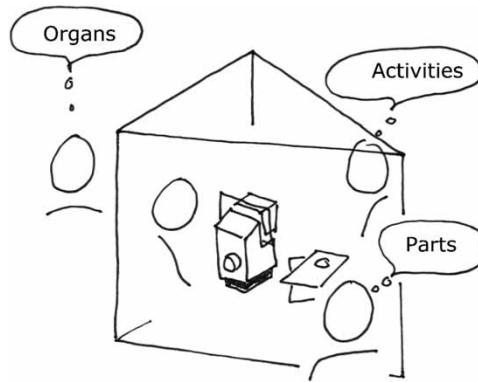


Figure 11. The product and its related use activities seen as three domains, in which the synthesis of the product may progress (Hansen and Andreassen 2002, Andreassen 2007).

As mentioned previously I see it as a mistake to bring in a function domain. The reasons are the following:

- Each domain shall contain a synthesis dimension in relation to the design task. Hubka's transformation model Figure 3 tells us that both transformation process, in my terminology activity, and the technical system shall be synthesised. The technical system needs two understandings or synthesis operations: how it functions, i.e. organ domain, and how it is built up, i.e. parts domain.
- In each domain we can reason backwards from wanted behaviour to structure as shown in Gero's model Section 3.5. In the activity domain we reason based upon operands: material, energy, information and biological objects. In the organ domain we reason from functions (effects), and in the part domain we reason from the part's tasks.

So from then on we used the illustration in Figure 11 to label the Domain Theory (Hansen and Andreassen 2002, Andreassen 2007).

3.4. *Mechatronics*

Danish industry recognised the mechatronic or multidisciplinary nature of their products early on, as the topic mechatronic emerged already in the 1970s. With the main purpose to establish and influence the teaching of design of mechatronic products (seeing control theory as being well established), the Danish Association of Mechatronics was established in 1977. I had the pleasure to join the establishing and many of the activities.

We established research in my group to clarify the following research questions, in a simplified form:

Does mechatronic design follow other patterns than Theory of Technical Systems and Domain Theory?

What are the characteristics of mechatronic conceptualisation?

Jacob Buur's thesis 'A Theoretical Approach to Mechatronics Design (Buur 1990)' showed the following results:

- Mechatronic systems comply with TTS and Domain Theory, i.e. they can be described in three systems views cross-boundary mechanical, electrical and electronic sub-systems.
- The synthesis can be treated cross disciplinary by functional reasoning.
- The nature of a mechatronic system follows the functions/means pattern (Hubka's 1st law)

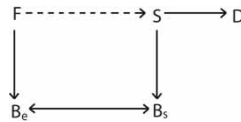


Figure 12. Gero's FBS Model (Function–Behaviour–Structure): Be, set of expected behaviour, Bs, set of actual behaviour, F, set of functions, S, Structure, D, design description, $-\rightarrow$, Transformation, $\rightarrow$, occasional transformation, $\leftrightarrow$, comparing (Gero 1990).

These contributions are central for understanding mechatronic conceptualisation. Buur showed that it is necessary to respect an effect oriented-definition of function, to recognise state transitions (obviously a topic neglected in mechanical engineering) and to introduce functions which establish the state transitions, when carrying out mechatronic conceptualisation.

Buur's results, especially the cross-disciplinary functional reasoning, are original, compared to several European attempts and to VDI 2206 (2004), where the conceptual design seems unsolved. It is not sufficient to let the disciplines unfold their partial models as long as these as a precondition have that a concept is formulated. Buur's research shows the power of a theoretical foundation like TTS and establishing necessary rigour in definitions.

3.5. Theory of Properties

The core nature of designing is captured in Gero's 'Function-Behaviour-Structure-Model' Figure 12 (Gero 1990) telling us, that we reason from required function to the product's expected behaviour, and jump to imagined, found or synthesised structures or solutions; the behaviour of these structures is compared with the expected, and when a reasonable identity is found, we face a good solution.

In Gero's model the concepts of function and behaviour are central. In industrial practice a list of desired functions and properties is used to guide the search for a good solution. In the light of the Domain Theory we have to ask the question of Gero's model: what design object(s) do we speak about? Also the activities related to the product? We see that as a challenging research question:

How to reason about expected behaviour?

Already in Tjalve's book we find a clear description of a product's characteristics, called the designer's degrees of freedom, and reflected in the methods of variation he allocates to the different design steps. This description does not apply to the activities related to the product, but Mortensen (2000) later worked out proposals for characteristics of structures in all three domains, as we shall see later.

Function can be seen as *what the product can do* and *what we can do with the product*, as long as we speak about desired function. So we are facing two different phenomena, both including the word function, but with two different meanings:

- The product composing or configuration activity, where we use a strict effect-oriented function concept for being able to reason about interactions between organs.
- The need/goal/use formulating activity and creative search for ideas, where we accept users' and customers' daily language and we play with the language in the ideation.

The first type of function concepts is the one which German language literature has been focusing upon and which has been taken up recently again in efforts in USA to clarify the nature of functions and function reasoning, see Stone and Wood (2000). The second type, which is related to use, purpose, utility, need and esteem, and which determines the purpose or 'the idea with the product' (see Section 3.10) is at present in the focus of design philosophers, see Crilly (2010).

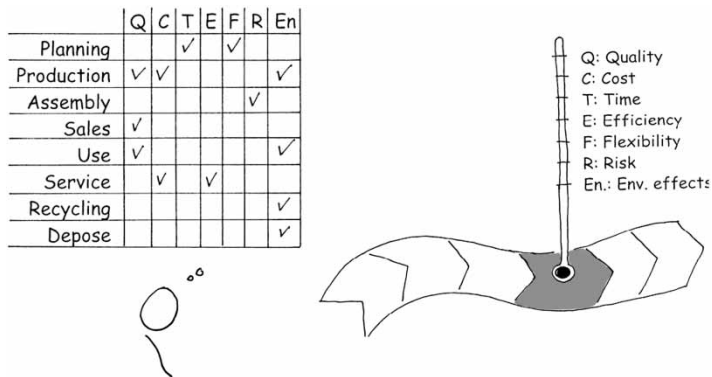


Figure 13. The properties related to an activity of the product's life phases, 'universal virtues', and the DFX-matrix linking virtues and life phases together (Olesen 1992).

The practicing designer shall show understanding of both phenomena and master the function reasoning in both.

When it comes to properties we see the same need for distinguishing users' imagination and perception of the product, and the designers' incorporation of properties. Figure 10 shows how we distinguish the following classes:

- *Eigen-properties* which are carried by the product in itself and can be observed without additional efforts.
- *Relational properties* which require a certain relation between the product and a situation to be established, so as manufacture, transport, emerging situations of risk, maintenance, recycling and re-use.
- *Allocated properties* that are required, in the minds of stakeholders and public such as 'pride of ownership', 'retro', 'Made in Germany', etc, allocated to the product and activities around the product and articulating wishes and values. Also functions may be allocated to a product, for instance tax authorities may see a product as a tax object.

We find relational properties in all DFX-dimensions. Olesen (1992) proposed a set of property classes related to activities as shown in Figure 13, namely quality, cost, time, efficiency, flexibility, risk, and environmental effects. He calls them 'universal virtues' for underlining the stakeholder's interest in the activity's proper performance. Together with the product's life phases as shown in Figure 13 we can formulate a DFX-matrix, onto which we can plot a distinct company's area of interest. In the matrix we can see how, for instance, design for assembly can be measured by cost and time, as mentioned before.

The soft non-engineering area of properties is a challenge in design, because the interpretation of user reactions and value for users are difficult to relate to concrete product properties. Olesen pointed to the difference in thinking pattern between quality and value as shown in Figure 14. Value is based upon the user's experience throughout the life cycle, expectations, social esteem and culture; Olesen used here the metaphoric picture of 'bicycle vs. bicycling'.

3.6. Goal statements

An important application of the Theory of Properties is the creation and use of design specifications or goal statements in the form of documents established early in a product development project and used for navigation and evaluations. Textbooks introduce specifications in a stereotypical

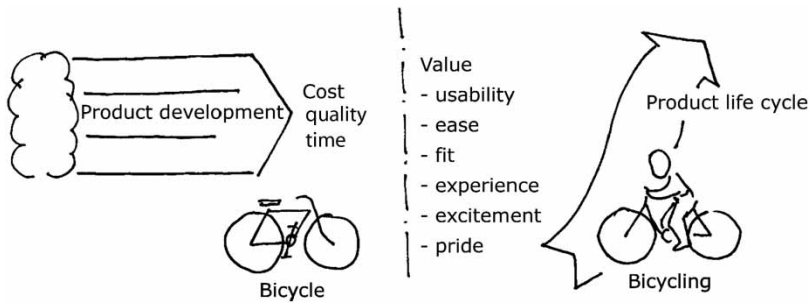


Figure 14. The difference between quality as part of the industrial credo 'cost, quality, time' and value is the actual user experience (Olesen 1998).

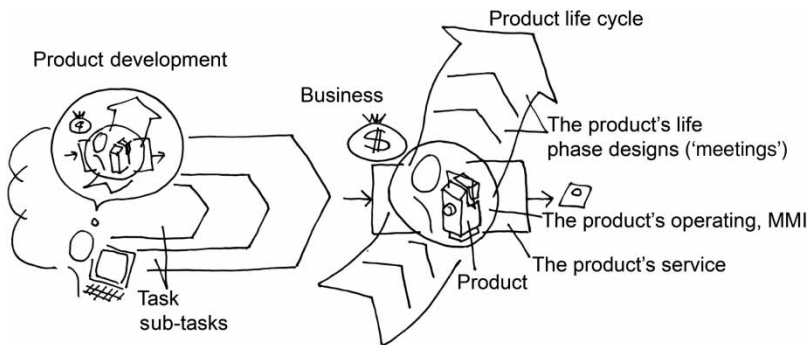


Figure 15. During the design process the designer has to lay attention in the decision making to several design objects as shown: the product, its use-related services, its fit to systems and life phase activities and the established new business (Hansen and Andreasen 2004).

way and there seems to be no theory behind the practice of formulating goal statements. We may articulate the research challenge as:

How can we best bring a goal statement to work? Can we articulate a goal statement supportive for ideation and decision making?

Our research has shown that several superimposed tasks to be delivered by the goal statement are expected from the users (Hansen and Andreasen 2004), but it is unclear how the document shall be filled in to support these tasks in the best way. The balance between articulating the wish for an innovative product and filling the document with specifications for an easily foreseen traditional product is a problem in practice. A closer investigation into the document's support for conceptual design shows that we only need a few, well articulated statements on values, important context aspects and key functions to get support for ideation (Hansen and Andreasen 2007). Investigations into decision making concerning concepts shows that we need to respect plural items in our decision making as shown in Figure 15 (Hansen and Andreasen 2004).

3.7. Reasoning about properties and quality

Galle (2008) points out that the most distinct competence of a designer is the ability to *predict* properties of a design, but also reasoning about properties to be articulated in a goal statement is demanding. We see problems in practice and education especially by treating 'soft' or 'distant' properties, which are not directly related to the product, and we see lack of understanding about

how desired properties can be related to the technical activity which may be performed by the product.

In many situations we see that certain properties are carried by other design items than the product (Hansen and Andreasen 2010). By a design item we mean an artefact which is fully synthesised or partially determined by the designer. Above we saw the items which were synthesised (the product, its services) and partially determined (business, product life systems like assembly and reuse). But some items or phenomena are not as evident and may easily be neglected. An overhead projector's ability to deliver 'high contrast' on the screen requires focus on the whole classroom setup, and a car's 'safety' can only be evaluated when the driving situation, the driver's abilities and the traffic situation are unfolded and mutual influences mapped.

In our research we have made investigations into how to reason about properties, first of all in the pattern of Design for Quality. Mørup (1993) established relationship between quality thinking and Hubka's Theory of Properties. He proposed a distinction between what the customer sees as qualities (for instance in Kano's sense (Kano *et al.* 1992)) and what is perceived as necessary efforts to establish a certain quality's level of performance, which he labelled as Q (big Q) customer quality and q (little q) quality efficiency, respectively. This interpretation brought clarity into the relations between design and production and was eagerly accepted and utilised by our industrial cooperators.

Hubka saw quality as the perceived and resulting evaluation of a product's properties. The maximal obtained quality is seen as ideal, desired value. Mørup's point of departure in his research on quality was the recognition of the partly subjective nature of quality and symbolic, emotional and social aspects of a product's value – plus the recognition that Design for Quality was in its infancy. In spite of TQM efforts the results of a quality focus in companies seemed sparse, when we empirically registered companies' confusions and disagreements about quality (Mørup 1993, McAloone and Andreasen 2001).

Based upon the Domain Theory, Mørup treated the relations between activity-, function-, organ-, and parts' characteristics and perceived quality. But his research also had substantial design methodology content. Mørup recommends eight elements of DFQ-efforts in a company, related to strategy, organisation, methods and especially a DFQ mindset. We believe that the notation of customer defined quality and the non-analytical relation between perceived value and designed properties is of highest importance for industry.

In Section 2.2 we told how Design for Assembly was leading to identification of a big selection of design principles. In a similar way we created contributions to Design for Reliability and Robustness through the thesis of Andersson (1996) and Matthiassen (1997).

Hubka's Theory of Properties became for us a rich and important area which supported us in creating design methodology to be brought into the education and practice and especially that the mindset of understanding a palette of properties and their different nature should be established.

3.8. Design for use and usability

In our research we have taken several steps away from the product's designable characteristics to properties like quality, value and allocated properties. As mentioned we see function both as what the product can do and what you can do with the product. From this there is a direct line to interaction. Tom Hede Markussen studied Interaction Design (Markussen 1995) in the 1990s based on the idea to identify the operational characteristics or the design degrees of freedom determining the quality of a product's use. He advised that an engineering-wise approach (grounded on causality based structures of the design object) and an experience-based approach (grounded in user's and designer's experience of, and reaction to, design) should be balanced. Markussen

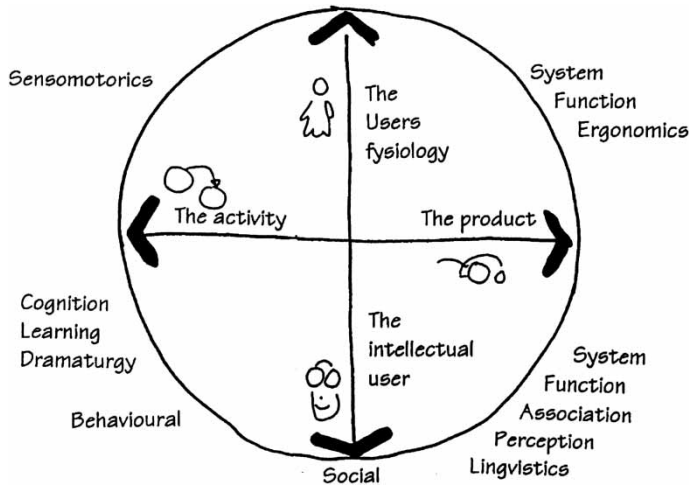


Figure 16. The unfolding of four dimensions of interaction and the different views, which can be used for a design strategy (Markussen 1995).

continued on the path that Tjalve created, adding a rich spectrum of prototype-related scenario techniques for ‘designing the use’, and covering a multitude of aspects of use and approaches to use design (Figure 16).

Many activities related to the product’s utilisation are influential for the product’s exploration and the user’s value experience: mounting, ready making, use activities etc. Pi Nielsen confronted in her thesis (Nielsen 1999) TTS with Human Computer Interaction and focused on the activity domain in the product’s use phase, but also on the physical product’s mediation and information related to the use. Nielsen points out that separate focus (by scenarios, experiments, prototyping) should be devoted to a product’s use activity, actions and operations, for the designers’ understanding of the mediation, sequence of operations, existing work practices’ influences, and to what degree use is as planned or situated. Similar to Mørup, her research pointed out our limited abilities to reason from the actual design to its qualities, including usability, unless we put the product in the hands of the user.

This research on the human and social side of product’s use and utility point to today’s strong industrial interest for user-based and user-oriented design and have given us a good understanding for teaching socio-technical aspects of design, see Section 5.6.

3.9. Design for environment

When we entered the design for environment area mid-1990s, dominated at that time by Life Cycle Assessment (LCA) methodology, the design practice was characterised by idealism and paralysed by the lack of understanding of synthesis. Olesen applied his theories of disposition and integration (Section 4.4) and created together with a local research group a guideline for practical design for environment (Olesen *et al.* 1996). The basic idea was to identify relations between product and life phase system characteristics (Figure 17), to understand what reasons for environmental effects are and to find potential mechanisms for reducing these effects. The mindset creating model Figure 18 shows how reasoning about so-called ‘meetings’, product and environmental effects may optimise the consideration of environmental condition during synthesis.

The strengths of the philosophy and methods are the fit into and balancing against already established procedural and organisational aspects, the pointing out of the need for real, actual, relevant product life insight instead of the normalised, ideal world of LCA, and the power of

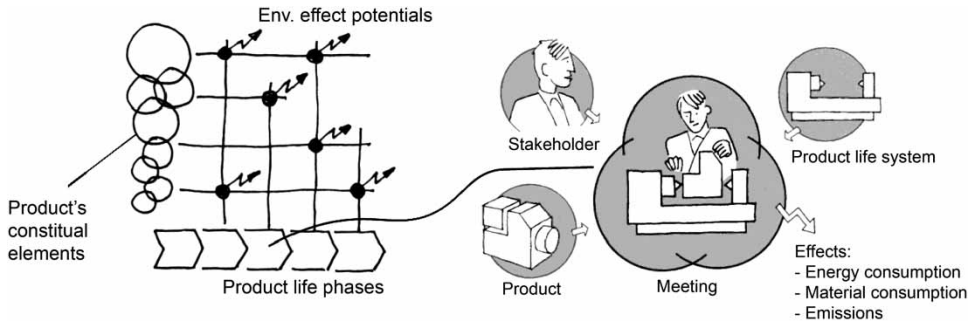


Figure 17. Environmental effects stem from 'meetings' between a product, a stakeholder and a product life phase system, and they are related to components of the product and life phase activities (Olesen *et al.* 1996).

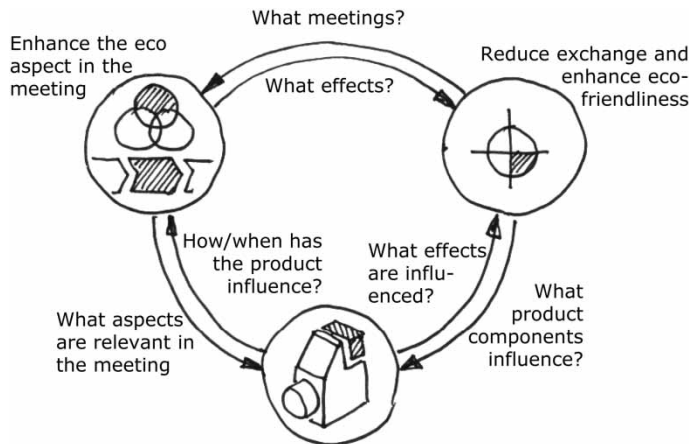


Figure 18. Mindset model for design for environment (Andreasen 2007 after Olesen *et al.* 1996).

visualisation of product life aspects and meetings through a so-called 'Product Life Gallery' technique, see Section 4.6.

3.10. Conceptualisation

Conceptualisation is the core of design synthesis. Our interest for conceptualisation stems not only from the growing interest from company managers to manage the innovative aspects of business, but also our wish to clarify what to tell the students about conceptualisation. Traditionally the creation of concepts was seen as a design phase, as something emerging from problem analysis, goal formulation and creation of (technical) ideas. The initiation of new product development comes from identified needs and opportunities and from ideas or necessities in the company's portfolio. It means that the concept should be *the answer* to these dimensions and therefore not only a matter of creating ideas.

So, a concept is a solution proposal described by such characteristics that we can see 'the difference that matters' compared with existing products, and we can see the answer. Hereby we overcome the paradox that we can continually create new concept cars, even if the car, so to say, was conceptualised hundred years ago: new things matter!

When an idea relates to a product, there should be two distinct dimensions in the idea (Hansen and Andreasen 2003):

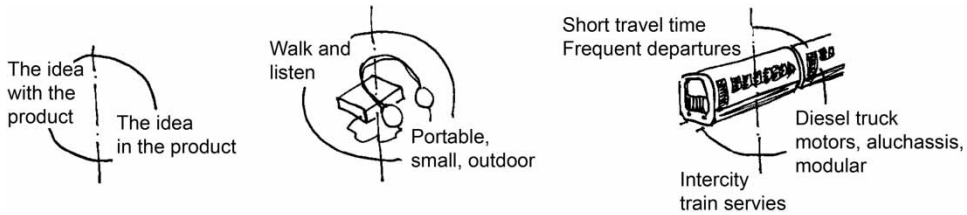


Figure 19. The two dimensions of an idea (Hansen and Andreasen 2003).

- The *idea in the product* seen as innovative functionality and new ways to realise these functions. This may be seen as the technical world.
- The *idea with the product* seen as innovative need satisfaction, and there lies the *raison d'être* of the product. This may be seen as the social world.

Figure 19 shows this mindset of ideation and some examples. We see mindset as an important challenge, especially for designers with an engineering background: 98% of all topics at our university aim at getting better 'ideas in the product'. Therefore engineers are not of high value when companies' strategy and portfolio shall be decided.

At present Claus Thorp Hansen and I are working on a textbook (Hansen and Andreasen 2010) on conceptualisation, based upon a line of topics including those mentioned above:

- (1) Establishing the insight which goes into a specific conceptualisation.
- (2) The ideation process creating concepts:
 - (a) Strategies for finding concepts
 - (b) Mental strategies or thinking patterns
 - (c) Use of graphical means
- (3) Evaluation and decision making
- (4) Presentation and argumentation for the best concept

Our picture of the reader is a designer who shall identify the *arena* on which the company and product shall operate, unfold creative, innovative patterns in a *mental space* by the team members, and being able to *stage and conduct* the conceptualisation activity.

Our effort draws together the basic aspects of Design Philosophy, Domain Theory, Theory of Properties and Design Methodology. We emphasise the creation of basic theoretical understanding of mindsets and practicing methods, especially the work practice related to methods.

4. Product development

The third dimension of my career is devoted to product development. Looking back, a new paradigm was created and broadly utilised by Danish industry, and we have improved our understanding of many development phenomena since. Today we face the challenge to create a next paradigm to benefit our teaching and industrial application. Let me explain the route.

Our teaching of design methodology in industry, our developing of industrial equipment at IPU (see Section 1), and working as consultants eventually showed us that models and methods belonging to mechanical engineering were of importance in the companies, but many aspects and goals in designing were out of focus in the designers' activities, even if they belonged to important business aspects for the company. At the same time we saw signals from other countries that product development should not only be seen as an engineering activity, but as a market/business/production-related activity of substantial importance for managing the company.

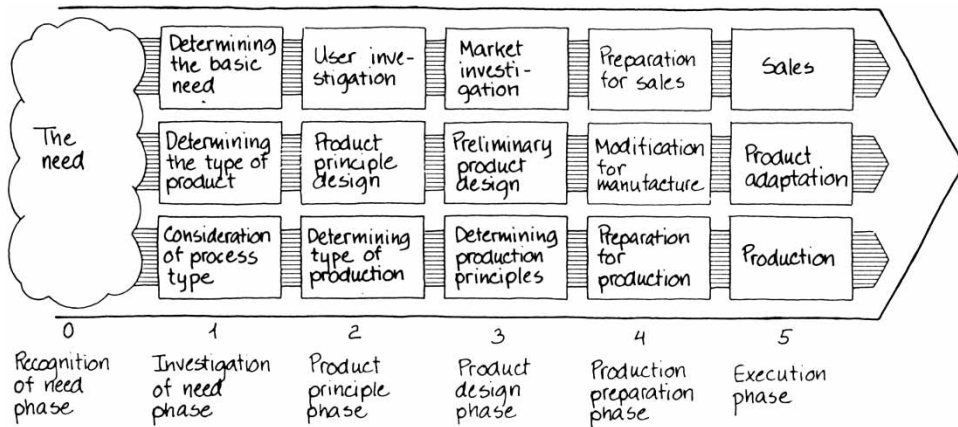


Figure 20. IPD, an ideal design model showing integration of market and production activities for creating new business (Andreasen and Hein 1987).

4.1. A need for integration: IPD model

Danish industry was very focused upon higher performance in new product development. As our contribution to industrial enhancement we formulated a research activity, financed by research grants and industry and planned to be launched as a broadly arranged campaign in industry. The main visible result was the book 'Integrated Product Development' (IPD) (Andreasen and Hein 1987). The text is formed as a series of essays telling about a typical company's product development activities, organisation, staff and behaviour in a mix of provocations and advisory statements. The core model of designing a procedure, is shown in Figure 20:

Today one would say that the results were created by a kind of participatory research and the research question in an after-rationalisation could be:

What is a comprehensive understanding (mindset) of industrial product development?

Key aspects of our model or better to say our procedure (ideal design model plus allocated methods) are (Boe and Hein 1999):

- We provide a model, which shows the context of engineering design. We see engineering design as a progression from a (management-formulated) goal statement to a complete production specification, while product development takes its start in a need situation and ends up with established production and sales, launching products that satisfy the need and becomes a new business.
- The model provides a generic map, defining the roles and activities of marketing, production and development, and shows optimal simultaneity and opportunity for integration with its pattern and milestones.
- The model gives words and graphics to some of the important phenomena governing product development, for instance the relations between allocated and consumed costs as a function of a projects lead time (Figure 21).

The effect of implementing IPD as procedure in a company was not only carried by the mentioned aspects, but also of creating business thinking, planning culture, competitor analysis, and utilising DFX-tools. In most companies it was also a necessity to create a well defined organisation regarding responsibilities.

IPD became state-of-the art in Danish industry and was vital for 10–15 years, before Lean was added or Cooper's stage-gate model was preferred. The main effects harvested by the companies

We observed a surprising phenomenon when we introduced IPD to companies. Normally we arranged a 2-day seminar with lectures and discussions alternating with statements on current problems and challenges in different function units like marketing, production, sales etc. Eventually a consensus about necessary improvements (and acceptance of IPD) grew out of the participants' talk, typically governed by a few influential persons in the organisation. Often this consensus was not controllable by management and often the seminar concluded with formulation of demanded changes to the management. What we naively believed was a traditional seminar, was in reality a social consensus operation (Andreasen and Hein 1998).

In our efforts to innovate in Danish companies by focusing upon the product development function, we joined, with influential consultants, a state-supported project with the goal to create a tool for change. Our research identified performance measurement tools, isolated patterns of 'illnesses' and 'cures', and in this way we could perform diagnosis and propose new patterns of goals, organisation and methods. The project resulted in a long line of interesting models and patterns, related to development tasks, job descriptions, team organisation, project strategies and patterns (Andreasen *et al.* 1989, Kirkegård 1989). Unfortunately the utilisation in industry became sparse due to industry's prevailed interest for soft motivation and behavioural approaches in the period.

4.4. Integrated production systems

A state financed research programme with contributions from two Danish universities was established in the mid-1990s together with industrial partners to develop new approaches to industrial integration. In our group we attacked the modelling problem and the general question:

How do product design decisions influence the product life activities?

The product life cycle from production via use to disposal is treated in the different DFX-areas (Section 2.2), but Olesen (1992) established a basic theory concerning the mutual dependencies between design solutions and the way a certain life phase activity can be performed, the Theory of Dispositions. To the vocabulary of Olesen belongs the metaphor *meetings*, i.e. the concept that in each life phase the product meets a new system, a new condition and new actors with certain tasks. Many of a product's properties, the so-called relational properties (Section 3.5) are realised in the meeting.

Olesen called the dependencies he was seeking for dispositions, i.e. the part of a decision made in one activity which affects the type, content, efficiency, and progress of activities within other functional areas (Figure 22). By this establishment of a language for activities, characteristics and properties, Olesen formulated a matrix of 'all DFX-areas' mentioned in Section 3.5, shown in Figure 22, and his theory of dispositions may be seen as a general theory of DFX, covering a mesh of product life activities and universal virtues.

Based on the clarification of dependencies between the product and production seen as activities and artefact systems, Olesen has been able to formulate general patterns for designing in a product life approach. He advises that designers' concerns should be concentrated at the conceptual stage of designing in the pattern of the so-called 'score model', an important mind-setting model (Figure 23).

4.5. Product life thinking and multi-product development

In the late-1990s an industrially financed research programme brought new life into our research activities, which broke up in two related streams (Andreasen *et al.* 2001):

- Product life thinking based upon DFX-research, especially design for environment (Section 3.9), research on Environmentally Conscious research brought in by Tim McAloone as new

staff member (Simon *et al.* 1998, McAlloone 2000), educational application of modelling of product life activities and challenges on teaching innovation.

- Multi-product development based upon our research on product modelling, modularisation, configuration systems, product life system modelling and alignment of value chain effects. As mentioned in Section 2.9, this area was taken over by Professor Niels Henrik Mortensen.

In the following the product life thinking dimension, so to say grounded by Jesper Olesen, will be treated in details.

4.6. Product life thinking practice

One can claim that there has always been a focus upon fitting the products to their life conditions, by early goal statements on life criteria, from DFX-efforts, and by product life data management. But product life thinking is, in essence, to reason in the appropriate way: how do we want to see manufacture, distribution, use and disposal of a product? The area ecology and

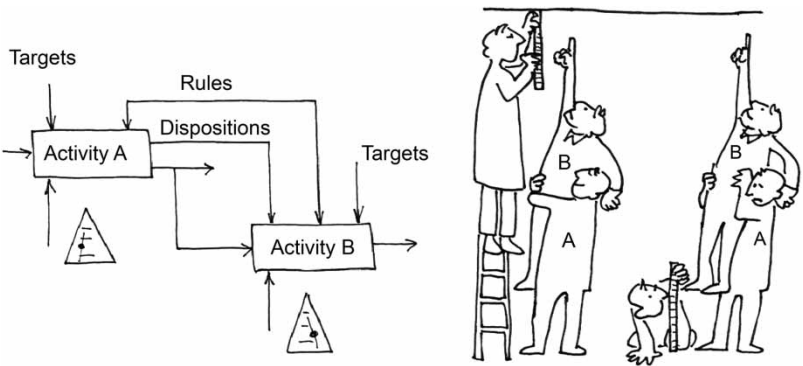


Figure 22. A general model of a disposition between two functional areas A and B. The cartoon illustrates the difference between being measured upon other's dispositions: the production manager B is measured upon what is actually the design manager's (A) result, – and to measure the disposition itself: how good is the design manager's dispositions in relation to production? (Andreasen *et al.* 1989, Olesen 1992).

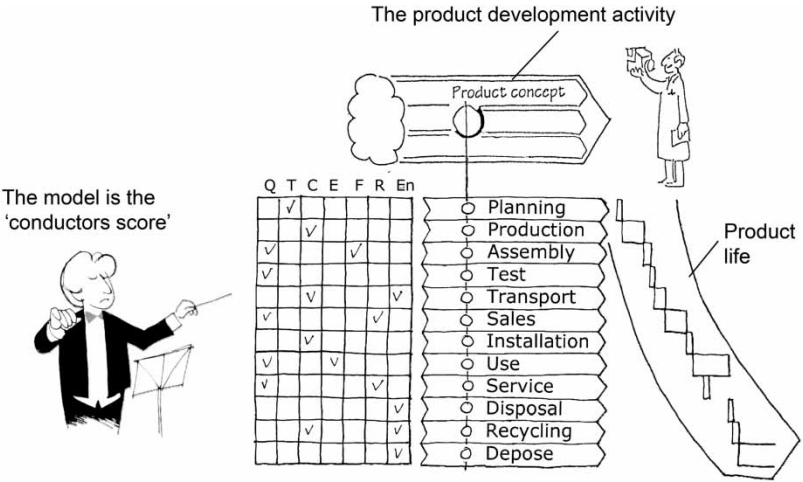


Figure 23. The score model as part of Olesen's product life approach (Olesen 1992).

consumption of natural resources has lead to eco-design theory and design for environment-methods (McAloone and Andreasen 2001). The following quotation captures the main idea of product life thinking: ‘For sustainable product development, it is essential, to first design total product life cycle in order to make reuse/recycling activities more visible and controllable, and then to design products appropriate, to be embedded in the life cycle’ (Kimura and Suzuki 1996).

A powerful instrument in product life designing is to make a product life gallery, to visualise on a series of posters, what happens in each life phase: what happens in the meeting? What do the actors have to say? The synthesis of the life model can take its point of departure in an existing product, afterwards changing into intended, desired and imagined ideal conditions for a new product. Each poster may tell important messages and be the seed of innovative ideas.

4.7. *Product/service-systems*

Product life thinking is bringing attention to the fact that the customers actually have the main interest in prolonging and securing the period where the product is able to serve them. To deliver service instead of the actual product has been known for years in certain branches such as aircraft, power plants or complex medical equipment, but for many companies it is a not utilised possibility to expand the business to products plus service. We call these synergetic deliverables a Product/Service-System (PSS).

From a research point of view it is interesting how a service and its production and delivery is designed and managed in an organisation. McAloone and Andreasen first treated PSS in terms of product development theory in early 2000s (McAloone and Andreasen 2002), and this activity has since increased to a number of research activities, including a chain of PhD projects, two of which are currently completed, by Matzen (2009) and Tan (2010). A central research question, treated by Matzen (2009) and Tan (2010), is:

What does an appropriate model for designing of PSS offerings look like?

What is sought after here is a model, prepared for synthesis of PSS, similar to the Domain Theory’s constitutional models; i.e. to determine characteristics of a PSS.

Literature studies (Tan *et al.* 2008) shows a very broad palette of service offering types, from delivering of *goods* (like helping materials), *non-goods* (like instructions, insurances, control etc.) and *man-power*. But it is a surprising discovery that any service seems related to the activities performed with the product, see Figure 3 in Section 2.3, Hubka’s transformation system model. A service is characterised by the *service channel*, challenging the necessary input/output and services related to the operators in Hubka’s model: technical systems, human operators, information and management. Executing the service happens in a *service activity*, and the business relation is a question of network and value enhancement.

A service’s performance follows the universal virtues introduced in Section 3.5 and service usability, experienced by the operators. Conceptualisation of new PSS offerings may be based upon analysing a product’s life cycle and identifying the service layer in certain operations, see Figure 24 (Matzen and Andreasen 2005). The services provided as the black part may be supply of materials, systematic maintenance, reference data base system, or ISO 9000 certification of the actual operation.

Service thinking adds in an interesting way to several other design aspects treated in this article, which moves the attention by designing from the product to ‘product plus more’, like life cycle, use, allocated properties, DFX-areas, actor network and now service offers. I see these dimensions as an expanded view on designing, which we hope to be able to integrate into a comprehensive treatment of conceptualisation (Hansen and Andreasen 2010).

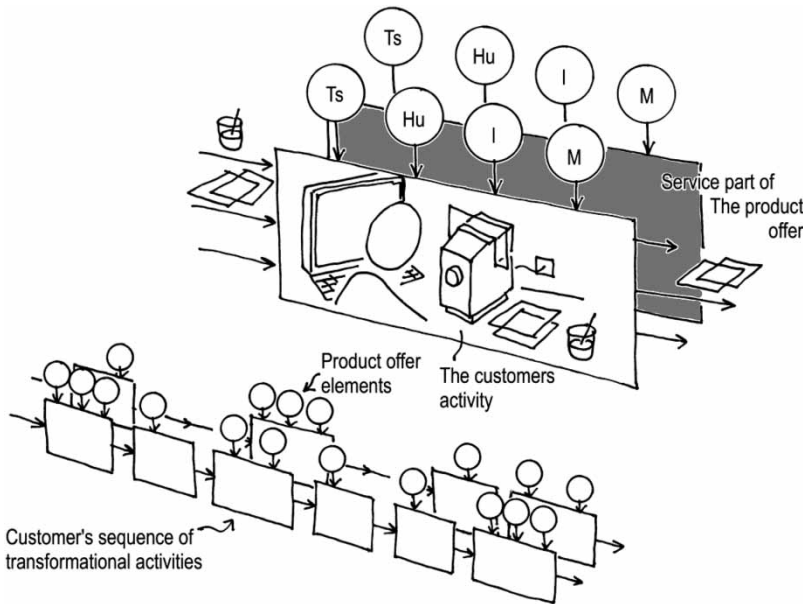


Figure 24. Service may be seen as an activity layer along the product's life activity chain (Matzen and Andreasen 2005).

I see the Product Life Thinking approach as a philosophy much more important for industrial innovation and future than for instance IPD (which is after all mainly a company-internal matter). The necessary totality of scope and integrity in what is created and delivered, from a single or networking company, needs to be based upon understanding of complex value chains, complex product/service operation, necessary complex network operation and meeting a composed, global market. The overall picture for such operation is sustainable results and agile, sustainable company operation.

Our research related to product development presented in this section started in a kind of participatory research where Lars Hein and I based upon our industrial experiences and dialogue with industrialists formulated what we saw as an explanation of industrial practice. But we gave our articulation a strong prescriptive form, which especially was meeting needs from marketing and production people: 'You showed us our proper role and position in product development' they claimed.

Further research mentioned above is a mix of empirical studies, participation in industrial activities, and theorising. An important driver has been industrialists' recognition that these concepts, models and methods empowered their daily operations.

5. My world view today

I see my personal development as a row of dislocations, as said in the introduction – situations occurred where my world picture cracked and I had to repair it – and a row of opportunistic situations, where ideas and concepts were taken up. I believe that my line of development to a high degree follows the time line of development of what we today call design science. In the following I will balance, draw implications and articulate perspectives in three balances concerning: Methods, What to tell the students? and Design Research. But first I introduce my context of operation.

5.1. *My workspace: WDK, ICED, DS*

On one of my visits to Vladimir Hubka in Zürich we created, together with Professor Umberto Pighini from Rome, the workshop and networking concept WDK, Workshop Design Konstruktion. Our ambition was to support the development of design methodology and one of our first initiatives was the ICED conference in Rome 1981. Hubka was already growing a network through visits, workshops and publications. The papers and attention were so promising, that we felt there was a good reason for continuing in a bi-annual conference pattern; the next was held in Copenhagen 1983, where I was co-organiser, together with Professor Christian Boe.

The ICED conferences were arranged by WDK and local organisers and supported by a group of notable professors and researchers, which met each year at Rigi in Switzerland. At the Rigi meeting in 2000 the international Design Society (DS) was founded with Professor Herbert Birkhofer as president and the ownership of the ICED conferences was transferred to DS.

The ICED conference in 2009 at Stanford University was number 17 in the series, and many formal ‘special interest groups’ (SIG’s) have been created. In the WDK-period we had also a high number of semi-formal and informal workshops and co-operations, from which the strong network of ICED participants- grew up. For me these networks have been the place where I captured a ‘first row’ insight into new areas and developments through arranging events, by reviewing papers and participating in the arrangements. And together with my colleagues we dared to deliver papers on new ideas and thoughts, which especially in the small groups led to very productive discussions.

Looking back, I see the WDK arrangements as a marvellous instrument for a researcher’s development, because of the interesting possibilities which were created. What actually were the secrets of the success is difficult to judge; Danish authors of a book on networking, which used WDK as a case study, wrote that it was the mission and the value concept of give-and-take which was the basis (Dalsgaard and Bendix 1996).

5.2. *The nature of methods*

We are methods makers, both my group and the main body of participants at the DS’s conferences. I am intrigued by the concept of methods, how they are learned and brought into practice. My PhD student Araujo (2001) investigated industry’s choice of design tools and created an interesting model of the designer’s understanding of procedural mode, i.e. how a task is perceived, interpreted, and executed based upon method knowledge and skill, and how results are brought into a contribution to clarification in the design process.

I have brought up the question of the evident difference between a method’s formal description and the necessary understanding for proper execution in a treatment of what I call *mindset*, understood as not only insight into the theory behind the method’s fundamental mechanisms, but also an understanding of its proper application (Andreasen 2003). Mindset has been mentioned in relation to Q/q, to dispositions, product life thinking, ‘the idea with/idea in’ concept, etc.

Most concern, also of Araujo, into the evident and unexpected modest use of design methods in industry, is related to the belief that the reasons are to be found in a method’s logical description, use of words and proper learning. I believe that design methods application in practice is much more delicate and a matter of social behaviours, negotiations and political forces.

Recently we have established a master course-module on design methods, which became another dislocation for me. It is based upon a socio-technical understanding of methods roughly articulated in the following statements:

- A method is a prescription or instruction of, how an actual task shall be done.
- Methods belong to a context which makes the actual application meaningful.

- Methods' execution builds upon an interpretation of the reality and the practice they shall operate into.
- Applying a method happens in a social system, a community of practice and is the result of negotiations, interpretations (especially of data) and evaluations.

The students apply empirical methods for investigating how an established method is used, its origin and agenda, and its interpretation by the actors. More than 50 reports have been delivered. The reports show that even if designers often attribute their results to methods (also in situations where it is obvious that the method is not doing the work), we can only get proper explanations of their use by understanding how the designers talk and feel about methods, individually and collectively (Jensen and Andreassen 2010), strongly conflicting with the claim of methods as being logical mechanisms. We have to understand carefully why and when they function in practice instead of seeing them as elegant, logical and indispensable deliverables which industry should not neglect.

How did we perform in our group concerning creation of methods? There is no doubt that we had our main attention upon creating a school (see Section 5.7) based upon a comprehensive collection of theories, models and concepts. We have been lucky to create tools which have had an industrial impact, even as we as typical toolmakers have neglected the tools' proper domestication. Today we face a challenge to supply the students with a rich understanding of designing, hopefully empowering their performance in industry.

5.3. Design research

Very often I realise that I do not really know what design science is. But I find the question very challenging and it makes me every year look forward to discussions at the summer school on engineering design research, which I run together with Professor Lucienne Blessing and Professor Christian Weber (Blessing and Andreassen 2005).

Duffy and Andreassen (1995) launched the idea, Figure 24, that in design research we study a 'reality' or practice of design and create a model of certain phenomena belonging to that reality. From this model we may formulate an information model and implement this in a computer system. Our models can all be fed back into practice and influence that practice. Tomiyama *et al.* (1989) claims that a model is based upon a theory about the phenomena we study. So certain theories may support us in the transformations in Figure 25.

What is actually a theory in the design area? A theory is seen as an *explanation* of a phenomenon, and a real theory should predict *behaviours* of the phenomenon. This is what we adopt from the basic natural sciences. Design is also a natural phenomenon, as Ullman underlines, but designers' behaviour, the process' progression and the results cannot be predicted. We have to find other virtues of design theories.

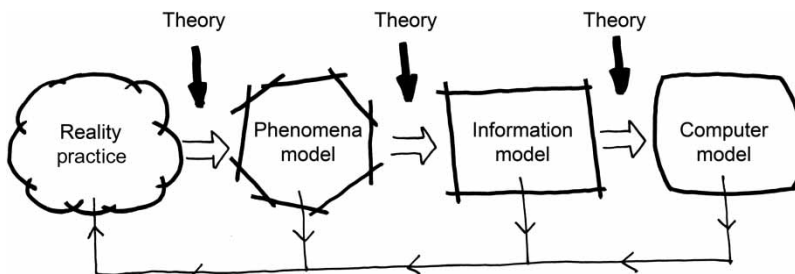


Figure 25. In design science we derive models from practice and develop tools and models for use in practice (Duffy and Andreassen 1995).

I believe that the most central behavioural characteristic of a design theory is that the theory leads to productive designing through the created mindset of the designer and the models, methods and tools; i.e. that it raises the *probability* of results and creates a space of solutions. You may say that this interpretation creates a diffuse link between a science and a practice, but it is generally agreed upon that the purpose of design science is to raise quality of designing and designs. This goal orientation is a quite unique dimension of our science, compared with the natural science goal: to create better predictions. 'Any design theory is worth precisely little until it has been applied and validated in practice', says Professor Wallace (personal conversations, 2010).

5.4. *Good research practice*

In an instruction for reviewing research programs I found the credo: 'Radical, Relevant, and Rigorous', to be used as criteria. Radical goes for two dimensions: of radical importance for practice, and radical contribution to existing theories. Relevant relates to industrial and application situations, including timeliness. And rigor covers the sharpness of research questions, use of concepts and theories, care concerning data, and a strong line of reasoning, which shows the results and their validity.

Defending a PhD is formally an education and has therefore two goals: a succeeded education in research craftsmanship and a research result. The project easily becomes a balancing problem between being an educational program (in Scandinavia time consuming formal courses) and research work, often long periods of reading and collecting data. But an often missing element is the planning for result: what creative effort or strategy can ensure or at least increase the probability of a (radical!) result?

In the researchers' long, lonely wanderings in the desert I try to re-establish mental health by asking them to sketch their results. Where do you in this month imagine we will end up? How do the results look like? Make a scenario of the use of your result! In this way guiding stars are created.

PhD students at our summer school (Blessing and Andreasen 2005) often state that they have the feeling of working in a vacuum. They feel left alone, and when they look upon former, finalised research, it only seems to add on the shelves; the world did not change. I regard discussions as the most powerful research method. I travel long distances to meet a good discussion partner. I often see papers and books where I make the diagnosis, that this author did not encounter a good critic in due time; and I often ask our PhD students, mirroring their faces daylong in the computer screen: why do you believe this is research?

The good research group is the one which harvests the PhD students results: by early application in teaching, by publications at conferences and in journals, and by publishing books which shows the greater lines and patterns, created by the researchers.

5.5. *The role of practice*

In an article balancing design research, Finger and Dixon (1989) claim that a good research institution is characterised by mastering the craftsmanship of research, based upon a solid theory foundation and mastering *best practice of designing*. The last criterion is very demanding; I believe that what research institutions can arrange is idealised, partial design situations see Section 5.7. Designers in industry perform work practice in a 'community of practice', which is the object of our design research. Our task as researchers is to study work practice based upon theories, hypothesis and research questions. Our findings may contribute to theories and design knowledge (verified singular and general insight into phenomena related to designs and designing).

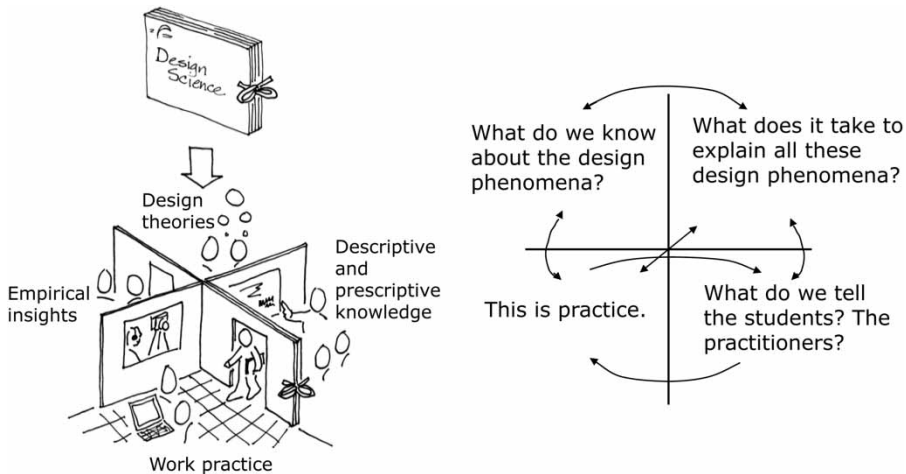


Figure 26. Design science seen as four interrelated worlds (Andreassen 2008).

Design Society's management is concerned about the so-called *consolidation* of design research, namely how to come to a common theoretical foundation and a respected set of models, methods and words. I believe one more dimension shall be added, namely the contributions to work practice. In an attempt to make my view visual I created the model in Figure 26 inspired from a children's book which opens into four rooms:

- The work practice which at the same time is our item of research and our 'customer area'.
- The empirical research dimension where we try to obtain knowledge about design phenomena.
- The design theory dimension where we try to structure design theories and create ontology.
- The dimension where empirical findings, design theories, models and methods are synthesised into a school of learning or production of literature for practice.

The arrows in the right-hand illustration tell that knowledge about practice is brought to practice from the three other 'rooms'.

Industrial practice contains endless different activities and operations we can study; from a research point of view, I believe that we are confronted with a much lesser number of design phenomena. Horváth (2004) proposes 39 topics in his structuring of design research.

The industrial practice and its industrial and societal role and conditions are changing dynamically, and the power of human endeavour may create surprising new directions influencing design research. What are the fix points of this? Design research seems to be little concerned about 'what' to design, focusing on 'how' and its efficient and effective performances. Thereby design research also seems to take pattern after needs in education and current practice: how to do it better?

I see a challenge in educating for change: candidates who dare to change industry, able to see new possibilities and mastering the staging of this industrial change. What they shall understand beside technology are the basic industrial, societal and human phenomena and they shall be able to make new instantiations, new design of design and use new creative patterns. Therefore I see the role of design research to deliver the patterns of 'how', and open to new dimensions of 'what'.

How does my picture of design match the research carried out in Copenhagen? Our activities have been governed by perceived industrial needs and the steps of progression we were able to take. Our results are interrelated by a common research foundation, namely the theories described in this article. In this way we may be seen as a school from a research point of view. We have not been aware of the possibility to identify and focus the research on identified classes of design phenomena, but have followed the general style of labelling our research with industrial terms.

In the research dimension of my activities I have met one of the strongest dislocations, namely the insight that designing can only be studied through multi-disciplinary research. Or articulated from the opposite direction: a pure engineering research background can only create insight into limited aspects of designing. Design research follows many research paradigms, differing from the natural science paradigm of repeated observations.

Through the years our research, similar to many other research schools, has met scepticism by university colleagues and in research funding councils, where research-funding proposals have been evaluated against ‘hard core engineering’ proposals on the one hand, or pure humanistic/social science proposals on the other. This situation is now fortunately beginning to change; there is a growing understanding of the role and importance of design research and increasing respect of the results. But we must continue to refine our research methods to match the four rooms of research, described in Figure 26, and dare to cross-disciplinary boundaries, in order to gain insight into the other important aspects of design research.

5.6. *Teaching design*

Teaching design, is in a simplified explanation, to transfer our knowledge about the nature of artefacts and how they are synthesised, and our insight into human behaviour, operation and cognition, to students. And to train the students’ abilities of imagination, awareness, ideation and foreseeing. The teaching is composed as ideally seen from several elements which have to be balanced:

- Introducing industrial practice and preparing students for industrial work. The engineering dimension is to know about technologies, industrial producing companies and market mechanisms. The design dimension is to know about the product development operations, their organisation, computer tools and management.
- Introducing the scientific foundation for a broad spectrum of areas: design theory, theory of technology and industrial development, socio-technical theory of technologies’ and products’ meaning and role in society, theory of work organisation, etc.
- Creating individuals with professional skills, understanding of societal, industrial and human individuals’ needs and values, reflection abilities concerning design work, own skills and attitudes, and life goals, entrepreneurial drive to industrial innovation, etc.

The teaching duties in my group grew throughout the years. The beginning was ‘engineering design teaching’, a mix of applied mechanics, design process insight and creative ideation. Later the link to machine elements was loosened and the elements design process, design practice, sketching and CAD, gained stronger profiles in differentiated courses. But our courses were ‘used’ in a mechanical engineering study line in a type of combinatory ‘picking from the shelves’, which did not give ideal conditions for our education goals strong dislocation occurred when we, together with teachers from socio-technical design created the idea of a new study line ‘Design and Innovation’ around 2000. Through a grass-root operation we gained permission for a separate intake of 60 students per year for a bachelor education and later for a master study. Teachers with industrial design background were added to the group. For me this was a realisation of an old dream, to be able to focus on a full composition as sketched above. Especially, I found it relieving now to be able to counterbalance the engineering ‘idea in’ thinking with a ‘idea with’ type of thinking, i.e. understanding and ability to unfold the meaning, role, value and importance of the social dimension of designing (Boelskifte and Jørgensen 2005, McAloone *et al.* 2007).

In the growth of design perspectives and development of new models and methods told above the rhetoric question ‘What to tell the students?’ comes up as the core question of relating research

to teaching. In the 1970s and early 1980s our main effort was to collect and select state-of-the-art insight into the curriculum. Later our research brought expansion into the teaching: product development, design for environment, conceptualisation, product life thinking, and innovation, and many new smaller elements were added. Newest developments on multi-product development and product service systems are brought in by the PhD students using master students as Guinea pigs. The understanding of research is expanded by building in empirical investigations and scientific reflections in several of the courses, which raises both the students' understanding of the scientific foundation of design and their interest in making a PhD.

5.7. Arranging practice

In a simplified pattern I see the more general insight into design as composed (Andreassen 2009) by:

- *Model-based theories*, i.e. theories on designing and designs based upon a mental model of artefacts and their synthesis, formulated by constructs of the scientist's mind, as mentioned in Section 3.3. Hubka's TTS, my Domain Theory, IPD, and Theory of Dispositions are such theories, and the same are theories of Gero, Dietrych, Koller, Roth, Pahl and Beitz, Suh, French and many others.
- *Ideas*, i.e. assumptions or postulates (and hereby following another meaning of the word theory). Examples are 'Form follows function' by Sullivan, 'The idea in/idea with' concept, 'Lean', 'The independence axiom' by Suh, 'Clarity, simplicity and safety' by Pahl and Beitz, etc.
- *Theories of design operations*, i.e. theories mainly based upon empirical studies which explain or describe design operations like cooperation, communication decision making etc.

Designing is, of course, also composed of theories belonging to the fields of natural sciences like mechanics, theory of machine elements, electronics, mechatronics etc. But these theories cannot explain designing and their proper application as such, nor does their content become the topic of design research.

How is practice developed in a company? Figure 27 proposes that it is composed by:

- Past experiences (following the company's path), industrial best practice and other cross-industrial influences.
- Industry's own perception of theories and ideas (often transferred by consultants).
- Influence carried by staff members' training and education.

These dimensions merge into a 'community of practice', i.e. the way a group functions.

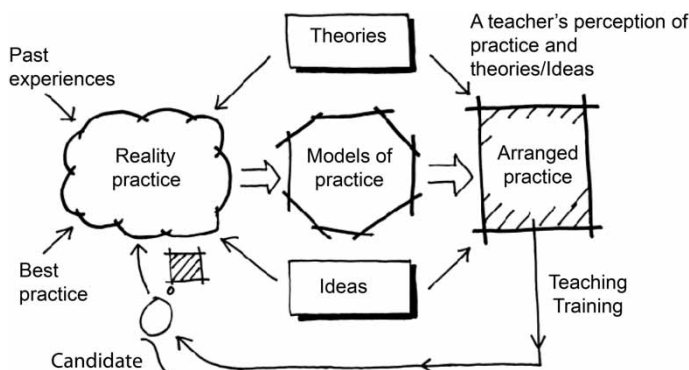


Figure 27. Shaping of industrial practice from theories, ideas and staff member's carried schooling (Andreassen 2009).

What the candidate brings into the company is experiences of teacher's 'arranged practice', i.e. the composed school of designing, combining lectured theories, ideas, methods and design approaches built into situations of skill development by different projects. The 'arranged practice' is the composed picture the students build up from the courses, each course giving a partial picture.

5.8. What then to tell the students?

A weak point in our new education has been the proper balancing or limitation of design dimensions to give space for the student's mastering of at least one technological area. The students have felt themselves unsafe when comparing with other educations at our university or when industry tries to bring them into known categories of engineers. We try to enforce the students understanding of their identity and ability by the label 'staging'. We believe the students' competences sum up in the ability to stage or conduct, to take responsibility for setting goals, planning, organising and managing composed, innovative projects. On the basis of their reflections, our education programme is now being adjusted.

It is our ambition that the education should be interesting also for students from electronics, mechanics, information technology, food technology etc., where the formal courses on designing are missing. But we have not yet reached the ideal structural flexibility within the programme or the university for managing that.

Design education is about learning how to think and do. We supply the students with concepts and models helping them to think, with projects helping them to do things and through lectures they shall understand and reflect upon the reality in which design takes place. At the moment we are curious to see the students' reflections: does our rather composed course pattern give a coordinated, comprehensive picture through our 'arranged practice'?

The perspective in our education is to empower the students for a future with the following set of characteristics (McAloon *et al.* 2007):

- Global activities spread in time, place and culture.
- A pairing activity, between technology- and market (user)-based.
- Built on individuals' abilities to act in teams, to network on many levels of organisations and in society.
- With a responsibility for self, company, society (Corporate Social Responsibility) and within the boundaries of sustainable development.
- Based upon the understanding of life-cycle – not only the product.
- Coupled with service as the main deliverable.
- With engineers being allowed into the decision process and development suite, *only* if they can prove themselves to understand context, complexity and business potential of their and other colleagues' actions.

We believe that our education programme is adding many of the aspects above and that our students' mindset and role identity match with this development. But we need to take a great deal of steps in a proactive and innovative direction in order to fulfil our ambitions.

6. Outlook

My article here is sketching the path through which I have tried to generate, based upon my world picture, a comprehensive model of designing, in what I call a school. The text is rather a sociology of design research than a gradual theory consolidation; therefore my text does not directly point

to what I see as challenges and open questions in design research and education. Let me make it short:

6.1. *Comprehensive understanding of designing*

Contacts with industrialists tell me that their imagination about design methodology is a complex patchwork with holes. Their practice brings fragments of formal understanding and methods into an often well functioning totality. I therefore see the need for modern textbooks in Pahl and Beitz-style linking engineering, Engineering Design and Product Development together, treating basic phenomena of designing and giving explanations to how to make it function in practice.

6.2. *The complexity problem*

Designers and design managers are facing complex ‘design machineries’ today whether it is strongly computer supported or based upon teamwork and cooperation. There are central dimensions in making them work: *Mindset dimensions* (The designers’ understanding of methods and ability to add what makes the methods function); *Usability dimensions* (making the machinery function in an innovative and productive way); *Socialisation dimension* (Staging the process and team work for creating a practice community); and *Utilisation* (to dare to utilise the machinery to its best performance).

When I see complex systems, for instance modular product family models and configuration support, I cannot help comparing them to the design of a church organ: very delicate and complex, created by people with different professional backgrounds. None of these will be playing the organ when it is finished. Who trains the organ player to virtuosity? Who trains the virtuoso for playing for instance ‘The modular product family business symphony’?

6.3. *Research consolidation*

I trust that design research has a central role in creating the future high performance, innovative and socially consolidating design methodology. I have been part of the debate about design science and design research, which gradually have found patterns supported by the Design Society’s conferences and groups. We need to agree upon concepts, theories, models and methods, which we believe are productive. We need to agree on what we see as good research practice in our very special science, characterised by multiple research paradigms. But one dimension puzzles me: where are the products in our research?

Hevner *et al.* (2004) point out in their article ‘Design Science in Information System Research’ that their area is composed by *behavioural science* (predicting human or organisational behaviour, when they shall design or operate information systems) and *design science*. The design science paradigm, they claim, seeks to extend the boundaries of human and organisational capabilities by creating new and innovative artefacts.

At the Rigi meeting 1996 (see Section 5.1) I made a presentation ‘World Class Design by World Class Methods’ (Araujo and Andreassen 1996). The core illustration is showed in Figure 28. The idea was to present quotations and viewpoints upon the world-class dimension and to critically analyse the current situation concerning world-class design, both seen as designing and designs as artefacts. It was interesting to realise that we in our research circles have much more focus on the design activity, its methods and rationalisation than on the design result: what is actually a good product? And we have surprisingly little focus upon understanding our role in society, for instance compared with the industrial design community.

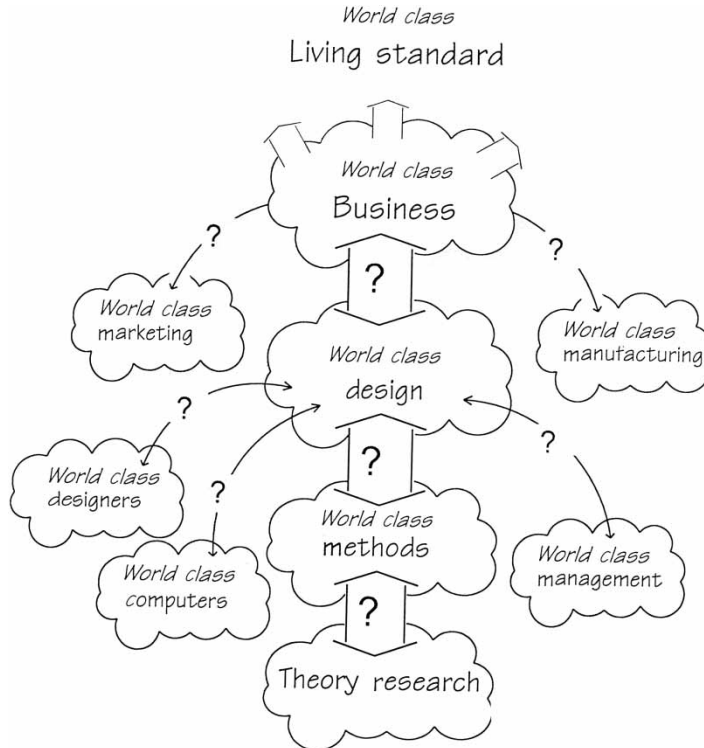


Figure 28. The chain from design research to contributions to living standard is long and easily makes our perspective as researchers blurred and egocentric (Araujo and Andreassen 1996).

In our design society we are tool-makers and there has been a remarkable movement away from mechanical and engineering design contributions to complex management and data management oriented papers at the ICED conferences. There is a long distance from such tools to innovative products to influential sustainable applications of the products in society. Have we cut our own lifeline?

Acknowledgements

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